Forward modules

Attention / MLP Module

$\mathbf{v}^{(1)}$ ..... $\mathbf{v}^{(K)}$

$\mathbf{e}^{(1)}$ $\mathbf{e}^{(2)}$ $\mathbf{e}^{(3)}$ ..... $\mathbf{e}^{(T)}$

Residual Connection transfer
updated special tokens

Backward and Descent modules

$\mathbf{v}^{(1)}$ ..... $\mathbf{v}^{(K)}$

$\mathbf{e}^{(1)}$ $\mathbf{e}^{(2)}$ $\mathbf{e}^{(3)}$ ..... $\mathbf{e}^{(T)}$

Attention / MLP Module (for Gradient Descent)

Attention / MLP Module (for Gradient propagation)

$\mathbf{v}^{(1)}$ ..... $\mathbf{v}^{(K)}$

$\mathbf{e}^{(1)}$ $\mathbf{e}^{(2)}$ $\mathbf{e}^{(3)}$ ..... $\mathbf{e}^{(T)}$

Residual Connection transfer
special tokens

Residual Connection transfer
forward module output

Forward modules

Attention / MLP Module

$\mathbf{v}^{(1)}$ ..... $\mathbf{v}^{(K)}$

$\mathbf{e}^{(1)}$ $\mathbf{e}^{(2)}$ $\mathbf{e}^{(3)}$ ..... $\mathbf{e}^{(T)}$